Yuyu Tang

First-year PhD Student | Computational Neuroscience

School of Physics and Astronomy, Shanghai Jiao Tong University

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Education

Shanghai Jiao Tong University | Current

PhD student, School of Physics and Astronomy | Computational neuroscience

Chongqing University

Bachelor's degree in Physics, Hongshen Honors School

GPA: 3.87/4.00 | Class rank: 2/105

Selected Research Experience

Real-Time Neural Decoding from High-Throughput Electrode Signals| Supervisor: Prof. Shanshan Qin

Undergraduate thesis in brain–computer interfaces | Chongqing University | 2026

（Chongqing Municipal Outstanding Undergraduate Thesis.）

- Benchmarked Kalman filter, MLP, and GRU decoders on three neural recording datasets, assessing within-session performance and generalization across recording days.
- Developed a hybrid GRU and cross-attention decoder with neural identity encoding, rotary positional embeddings (RoPE), and dynamic channel dropout to address neural signal nonstationarity.
- Achieved an R^2 of 0.336 in the cross-session M1 benchmark, compared with −0.799 for Kalman filtering, −0.751 for MLP, and −0.727 for GRU.

Deep Learning for Electric Field Sensing with Rydberg Atoms

Core team member | Nov 2023 – Nov 2024

- Simulated electromagnetically induced transparency spectra and generated datasets in MATLAB; built and trained fully connected neural networks in Python to improve electric field measurement accuracy.

Traffic Flow Prediction with Spatiotemporal Neural Networks

Core team member | Project initiated Nov 2024

- Implemented algorithmic components using spatiotemporal graph convolutional networks, LSTM networks, and clustering for traffic flow prediction and traffic state assessment.

Technical Skills

Programming and tools: Python, MATLAB, C++, PyTorch, scikit-learn, Mathematica, Linux.

Methods: Neural networks, supervised learning, clustering, numerical simulation, and data visualization.

English: CET-6, 506.

Academic Experience

- **Computational Neuroscience Summer School**, Westlake University, August 2026.
- **Japan Summer Study Camp**, Hongshen Honors School, 2024. Attended AI courses at the University of Tokyo and Waseda University.

Selected Honors

National Scholarship; National Encouragement Scholarship; university scholarships (6 awards).

Meritorious Winner, Mathematical Contest in Modeling (2024); Provincial First Prize, 15th National College Students' Mathematics Competition.